

Ten Layers of the First-Alias Frontier: Exact Transfer Architecture and an Inactive Actual Bridge

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July 2026

Abstract

RH-322–RH-330 pass from a certified folded Gaussian row to an exact first-alias full-trace transfer criterion. The chain contains local physical theorems, affine and exchange models, conditional composition and matching criteria, and a validated negative result for one frozen graded model. We audit these statements in their original data types. On a clock with $H_k = kR^{-2k}$, the existing identities reduce the actual first-alias constituent to

$$E_{\text{prefix}} = E_{\text{off}} + \frac{|e_k^a|}{2H_k}, \quad e_k^a = \widehat{e}_k + \Theta_k,$$

where $\Theta_k = \Delta B_k + \Delta S_k + \Delta R_k + \Delta P_k - \Delta A_k$. The first equality still assumes identification of the actual critical coefficient. A closing model therefore transfers exactly when $\Theta_k = o(H_k)$, but the identification, physical two-channel Duhamel enclosures, parity/alias replacement, signed far remainder, off-alias background, and determinant gluing are all unproved. The exchange and observation split is gauge-dependent; only their observable shell sum may be compared intrinsically. Thus the route coordinate is `first_alias_transfer_criterion_exact_actual_replacement_open`. This is a scoped route stop, not a full-trace theorem, and no Riemann-hypothesis conclusion follows.

1 Scope and common scale

Let $u_c \in (1, 2)$ be the root of $u^3 - 2u^2 + 2u - 2 = 0$ and put $\lambda = 2u_c(u_c - 1)$. As $\sigma \downarrow 0$, fix the moving first-alias order and phase by

$$k = k_\sigma = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad \eta_\sigma := k_\sigma - \frac{\log(1/\sigma)}{2 \log \lambda} \longrightarrow \eta, \quad H_k = kR^{-2k}, \quad R = 1.4.$$

The first five papers below establish local profiles and exact algebraic interfaces. The next four introduce the localized shell, its conditional matching equation, a frozen isolated model, and the actual/model transfer identity. None identifies the whole chain with the actual noisy full-trace coefficient.

paper	proved conclusion	data type	inactive interface
322	folded-row L^1 identity and phase stability	one physical row	paired trace and shell
323	oriented affine (V, U, W) law and L^1 isometry	local affine model	physical two-leg kernel
324	sharp positive $O(\sigma)$ first-leg remainder	one physical leg	second leg and composition
325	transported Duhamel criteria and two obstructions	conditional path/operator theorem	all physical leg and trace bounds
326	exact Hardy parity/alias identity and signs	algebraic packet	probability-to-trace map
327	actual localized partition; shell nonidentifiability	typed trace identity plus synthetic model	physical shell and far bound
328	matching equation, precision law, false-positive example	conditional exchange coordinate	identified physical fields
329	frozen rational model fails at scale H_k	graded isolated model	actual-operator replacement
330	exact signed transfer and grouped enclosure	conditional actual/model pair	physical signed hypotheses
331	typed synthesis and route firewall	review proposition	all actual bridge obligations

2 Exact observable interface

RH-326 and RH-327 give the five-slot sign convention

$$e_k = B_k + S_k + R_k + P_k - A_k. \quad (1)$$

Here B_k is the localized boundary slot, S_k the neighboring observable shell, R_k the signed far slot, P_k the even parity packet, and A_k the first-alias packet. For an actual packet and a frozen model, put

$$\Delta Y_k = Y_k^a - \hat{Y}_k \quad (Y \in \{B, S, R, P, A\}), \quad \Theta_k = \Delta B_k + \Delta S_k + \Delta R_k + \Delta P_k - \Delta A_k. \quad (2)$$

Then subtraction gives the exact identity

$$e_k^a = \hat{e}_k + \Theta_k. \quad (3)$$

The proposed decomposition $\hat{S}_k = \hat{X}_k + \hat{E}_{\text{obs},k}$ is not intrinsic. For every t_k ,

$$\hat{X}_k \mapsto \hat{X}_k + t_k, \quad \hat{E}_{\text{obs},k} \mapsto \hat{E}_{\text{obs},k} - t_k \quad (4)$$

leaves $\hat{S}_k$, $\hat{e}_k$, and (3) invariant. The review therefore audits ΔS_k and never claims separate exchange and observation accuracy unless a physical identification has first fixed the split.

3 Conditional closure theorem

Let $q_{\sigma,k,n}$ be the typed full-trace coefficient error. Choose $2k < h_\sigma \leq 4k$ and define

$$E_\sigma^{(h)}(R) = \sum_{2 \leq n < h_\sigma} \frac{|q_{\sigma,k,n}| R^n}{n}, \quad E_{\sigma,\text{off}}^{(h)}(R) = \sum_{\substack{2 \leq n < h_\sigma \\ n \neq 2k}} \frac{|q_{\sigma,k,n}| R^n}{n}. \quad (5)$$

Theorem 3.1 (typed first-alias transfer). *Assume the actual critical coefficient is identified by $q_{\sigma,k,2k} = e_k^a$. Then*

$$E_{\sigma}^{(h)}(R) = E_{\sigma,\text{off}}^{(h)}(R) + \frac{|e_k^a|}{2H_k}. \quad (6)$$

Together with (3), this gives

$$E_{\sigma}^{(h)}(R) \longrightarrow 0 \iff E_{\sigma,\text{off}}^{(h)}(R) \longrightarrow 0 \quad \text{and} \quad \Theta_k = -\widehat{e}_k + o(H_k). \quad (7)$$

If $\widehat{e}_k = o(H_k)$, the second condition is exactly $\Theta_k = o(H_k)$.

Proof. Separate the $n = 2k$ term in (5). Since $R^{2k}/(2k) = 1/(2H_k)$, (6) follows. Both terms on its right are nonnegative, so their sum vanishes exactly when both vanish. Substitution of (3) gives (7) and its closing-model specialization. $\square$

The coefficient identification in the theorem is an assumption, not an output of RH-322–RH-330. The theorem also leaves the off-alias term and the separate head/counterloop determinant budget untouched.

4 Why local success does not decide the actual trace

RH-325 retains transported incoming laws in every hybrid term. RH-330 retains all $4k$ terms from the two length- $2k$ critical channels before placing their signed sum into groups. This is the correct conditional architecture, but no physical signed group enclosure has been supplied.

Proposition 4.1 (same unsigned data, opposite critical verdicts). *Let $A_k/H_k \rightarrow \infty$ and let a comparison model close. Two replacement ledgers can have identical componentwise unsigned bounds and all other slots equal, while one closes the critical packet and the other fails there at divergent scale.*

Proof. Take respectively

$$(\Delta B_k, \Delta S_k) = (A_k, -A_k), \quad (\Delta B_k, \Delta S_k) = (A_k, A_k), \quad (8)$$

and set the other defects to zero. Both ledgers have unsigned budget $2A_k$. Their signed defects are 0 and $2A_k$, respectively. Equation (3) gives critical-packet closure in the first completion and critical-packet failure in the second. These are abstract typed ledgers, not claims about the physical operator; a weighted-prefix verdict would additionally require the common hypothesis $E_{\sigma,\text{off}}^{(h)}(R) \rightarrow 0$. $\square$

For the failed RH-329 model,

$$\frac{\widehat{e}_k}{A_k} \longrightarrow -(1 - C_* C_M), \quad \frac{A_k}{H_k} \longrightarrow \infty. \quad (9)$$

Actual closure would therefore require the tuned repair $\Theta_k = -\widehat{e}_k + o(H_k)$. Conversely, $\Theta_k = o(A_k)$ would conditionally transfer its negative divergence. Neither actual estimate is known. The scalar repair in RH-330 proves only that an isolated-model failure cannot itself decide the actual verdict.

5 Layer audit and inactive hypotheses

The local chain is internally consistent with the following strict scopes. RH-322 retains the clearance phase. RH-323 is an affine probability lift, and RH-324 controls one physical leg only. RH-325 proves a composable criterion but not its physical hypotheses. RH-326 fixes the parity and alias signs, while its scalar balance decimals are not physical interval certificates. RH-327 defines the actual localized slots; its inverse-branch clearance asymptotic is an inherited interface, and its exchange

completion is synthetic. RH-328 is a fixed-reference conditional equation. RH-329 is a graded family rather than one operator realizing every order. RH-330 is exact for a typed actual/model pair but does not provide that pair for the noisy operator.

The following actual obligations are all open:

1. identification of $q_{\sigma,k,2k}$ with the five-slot packet and of the physical packet with a frozen model;
2. the second critical leg, transported phases, and physical signed two-channel Duhamel enclosures;
3. target-scale parity/alias replacement and a signed far-remainder theorem;
4. vanishing of the off-alias weighted background;
5. the independent head/counterloop budget required for determinant gluing.

Their coordinatewise conjunction is not implied by the local or synthetic results. Hence the exact criterion is inactive.

Proposition 5.1 (abstract typed-completion underdetermination). *Within the abstract typed-completion class consisting of the RH-322–RH-330 ledger conclusions and no additional actual bridge hypothesis, neither $e_k^a = o(H_k)$ nor $|e_k^a|/H_k \rightarrow \infty$ is logically determined. After critical-coefficient identification, the minimal signed datum deciding the critical packet is $\Theta_k = -\widehat{e}_k + o(H_k)$; weighted-prefix closure additionally needs $E_{\sigma,\text{off}}^{(h)}(R) \rightarrow 0$, and determinant gluing additionally needs its independent head/counterloop budget.*

Proof. The two ledgers in Proposition 4.1 preserve the same componentwise unsigned information and every earlier local or model conclusion but have opposite critical verdicts. Thus those conclusions determine neither alternative. The signed condition is the equivalence in Theorem 3.1. The two additional requirements follow respectively from the nonnegative decomposition (6) and from the fact that (6) contains no head/counterloop determinant term. $\square$

This proposition constructs no physical noisy operator and does not assert that both abstract completions are physically realizable. Its negative content is strictly logical: the available typed ledger and unsigned data do not imply an actual verdict without a physical signed replacement theorem.

6 Executable audit and claim boundary

The executable ledger normalizes the heterogeneous Gate labels in RH-322–RH-330, records ten scoped conclusions and zero discharged actual bridge obligations, and keeps every bridge obligation false. RH-330’s six rational reproduction rows at $k = 2, 4, 8, 16, 24, 32$ retain 344 synthetic Duhamel terms. These rows test formulas and signs; they are not evidence for physical asymptotics. The publication audit hashes each paper separately and the whole ten-paper batch, excluding caches and L^AT_EX intermediates.

In the coordinate order (head, bridge, tail, target, boundary), the inherited typed branch ledgers remain

$$(1, 0, 1, 1, 1), \quad (1, 1, 0, 1, 1), \quad (10)$$

both score four, with no weighted cross-branch glue and complete count zero. Gate A remains open, so no canonical intrinsic dynamical spectral determinant is identified. Gates B–E also remain open. This batch constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.